Jason Bowman

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SUMMARY

Staff/Principal-level engineer – by turns hacker, maker, and academic – with over a decade spanning infrastructure, data, backend services, and open-source frameworks, from startup to IPO. The arc runs from founding Uber's Site Reliability Engineering organization and building out one of the era's largest Kafka deployments – paving the way for others in the industry to follow – through architecting, leading, and building Unity's company-wide data platform at millions of events per second, to cross-region payments infrastructure at Highnote. Years on the front lines forged one conviction, held across every domain: that correctness, resilience, and integrity are designed in – correct by construction, not by prayer and runbook – on computer-science fundamentals rather than guesswork. *Denful*, built full-time and in the open, is the pursuit of exactly that: composable, verifiable foundations that make rigor the default rather than a luxury, from a single laptop to a fleet of thousands. The work spans the full stack – from the Linux kernel and network stack, through distributed data platforms (Kafka, Spanner), to DSLs, type systems, and effect systems – and that rigor matters more than ever as agentic coding makes plausible-but-wrong systems cheap to produce and expensive to validate.

EDUCATION

The University of Alabama	Tuscaloosa, AL
Master of Science in Computer Science (MSc)	*2012*
Bachelor of Science in Computer Science (BSc), Minor in Mathematics	*2011*

SKILLS

Languages & Platforms: Scala, Nix, Python, Go, Java, C/C++, Ruby, and UNIX shell; 25+ years on GNU/Linux and UNIX; strong functional-programming background
Language Engineering & Tooling: DSL design, type systems, algebraic effects, term and graph rewriting, AST-based codemods and automated migration, parser generators (ANTLR, tree-sitter), macros and metaprogramming; sbt, Bloop, Metals, JMH, Gatling
Distributed Systems: Kafka, stream processing, high-QPS services in hyper-growth environments, elastic compute, API design, performance optimization, graceful fault detection and recovery
Cloud & Data: GCP and AWS – BigQuery, BigTable, Spanner, Dataflow / Apache Beam / Scio for specialized real-time pipelines, Avro (avro4s), GKE; PostgreSQL, Redis, ZooKeeper, Hadoop, Vertica
Systems: Linux kernel and kernel-module development, device drivers, network-stack and protocol optimization, source-level performance tuning, multimedia (gstreamer, ffmpeg)
AI & Machine Learning: Reinforcement learning (PPO, SAC, GAIL) and distributed RL training platforms (pre-LLM-boom); LLM agentic workflows, multi-agent orchestration, and spec-driven development
Kubernetes & Cloud-Native: Cilium (eBPF CNI and network policies), microVMs, hardware virtualization (VFIO, GPU passthrough)
GitOps & Infrastructure-as-Code: ArgoCD, Flux, Terraform, Ansible, Puppet, Chef, SaltStack, bare-metal and cloud fleet management, declarative secrets (agenix, SOPS), CI/CD and developer-experience tooling
Observability & Reliability: Grafana, Prometheus, Grafana Alloy, metrics and log pipelines, alerting; 24/7 on-call, automated incident detection and remediation, backup and disaster recovery
Security & Compliance: security auditing, applied cryptography (KMS envelope encryption, AES-CBC, key derivation), OIDC and access control, PII/GDPR data governance, HIPAA compliance
Leadership: Technical Project Management, Team Leadership, Mentorship, Organizational Leadership

EXPERIENCE

Denful Open Source
Co-Founder *2023 - Present*

- Co-founded an open-source organization building composable, aspect-oriented Nix tooling, grounded in algebraic effects, interaction nets, and graph rewriting, with an active and growing community (*den* at 450+ GitHub stars).
- **gen:** Authored *gen*, the ecosystem's foundational layer: a suite of dependency-light, pure-Nix libraries spanning algebraic primitives, higher-order scope graphs, typed registries, selector and graph-query algebras, and guarded graph rewriting.
- **den:** Lead developer of *den*, a context-aware, aspect-oriented configuration framework that produces NixOS, Darwin, and Home-Manager outputs from a single declarative source, using algebraic effects to make resolution explicit and testable.
- **Domain-Agnostic Configuration:** Generalized the engine beyond NixOS to arbitrary configuration targets, demonstrated by extending it to Kubernetes GitOps through upstream contributions to *nixidy*: a generic object match/rewrite rule engine and CRD-native module accessors.
- **Graph-Based Infrastructure:** Modeled service topology, firewall rules, and access controls as derived properties of a typed relationship graph, synthesizing network policies and ACLs from declared service relationships rather than hand-maintaining configuration.

Highnote San Francisco, CA
Senior Software Engineer *2022 - 2023*

- Infrastructure engineering for a B2B card-issuing fintech, focused on data systems and large-scale platform migrations.
- **Spanner Replication System:** Implemented a transaction-log replay and mirroring system on Google Dataflow for cross-region Spanner replication, sustaining 99.99% uptime for payment processing.
- **Codemods & Migration Tooling:** Engineered AST-based codemods and automated migration utilities driving mono-repo-wide framework upgrades across 50+ microservices, cutting manual effort ~80% and eliminating version skew.
- Delivered data pipeline infrastructure for real-time payment analytics and fraud detection, processing millions of transactions daily with sub-100ms latency requirements.

Unity Technologies San Francisco, CA
Senior Software Engineer / Tech Lead *2018 - 2022*

- Led engineering teams and architected large-scale data platform infrastructure serving analytics, advertising, and ML workloads at one of the world's largest mobile ad networks.
- **Data Platform (Sole Architect):** Architect and primary engineer of Unity's company-wide event-ingestion platform – the production replacement for a legacy daily-batch flow, built during the AWS-to-GCP migration while leading a team of 6. On Apache Beam/Scio and Dataflow (Kafka → schema'd Avro → GCS → BigQuery), it sustained 1M events/sec (10B/day) and cut end-to-end data-availability latency from 24+ hours to under 15 minutes at 99.99% availability, consolidating three previously independent pipelines (analytics, advertising, CDP) into a single platform adopted across the company.
- **GDPR & CCPA Compliance:** Architected the platform's privacy layer and led the cross-functional rollout with Legal and company-wide data owners. PII was split into a separate KMS-encrypted stream and access-controlled store, with subject-level cryptographic compliance keys enabling right-to-erasure across the data lake – plus PII hashing, geo-coordinate fuzzing, GeoIP enrichment, and automated statistical PII classification – flagging fields by value cardinality and uniqueness for data-owner review and refinement. Per-owner bucketing provided tenant-scoped access controls and cost chargeback.

- **Runtime Control Plane:** Designed a live control plane – BigTable-backed schema, routing, and encoding registries hot-reloaded into running pipelines – so data owners onboarded new event types and evolved schemas with zero redeploys, decoupling platform operations from per-team data changes. Automated schema derivation inferred Avro contracts from existing data.
- **Reliability Tier:** Built the platform's fault-tolerance layer – dead-letter capture/replay with rate-limited backoff and BigQuery load-job monitoring with automatic retry – sustaining delivery guarantees at 10B events/day.
- **ML-Agents RL Platform:** Architected a cloud-native reinforcement-learning platform on GKE for Unity's ML-Agents – applied RL (PPO, SAC, GAIL) years before the current AI boom – orchestrating distributed training experiments and Unity simulation builds, and scaling researchers' workloads across hundreds of nodes.
- **Unity Simulation & Robotics:** Stood up remote-tunneling infrastructure enabling secure access to Unity instances in shared cloud environments. Developed tooling to bridge on-premises robotics workflows with cloud-based simulation.

Uber Technologies, Inc San Francisco, CA
Software Engineer *2015 - 2018*

Marketplace Engineering // Fares *2016 - 2018*

- Led development of end-user-facing product features spanning back-end services and mobile applications.
- **Pricing Integrity Engine:** Led development of a real-time and batch data-validation platform for fare data, with configurable business rules, analytics, and automatic remediation.
- **Fares Platform:** Designed next-generation fare calculation platform for all Uber products using a descriptive DSL, integrating tolls, maps, surge, and subscription data.
- **Promotions Platform:** Created auditing and remediation tools for post-transaction analysis of user promotions across real-time systems.

Site Reliability Engineering *2015 - 2016*

- An early engineer at Uber and founding engineer of its Site Reliability Engineering organization. Originally served all of Data – the better part of a year on production-critical Vertica and data-warehouse reliability (30% of Uber's hardware footprint), automating provisioning, replication, ETL, and cluster management across Vertica, Hadoop, Redis, and MemSQL. Then, as a lead on the Streaming SRE team, scaled one of the world's largest Kafka deployments of its era to thousands of brokers at >500 GiB/s across multiple regions.
- **Kafka Infrastructure:** Built an intelligent partition-redistribution engine spanning hundreds of brokers across many clusters, driving load-distribution standard deviation from ∼53% to 0.0001% of mean, eliminating disk hotspots and self-healing broker failures. Kernel, filesystem, JVM, and broker-source-code tuning delivered over 5× (500%) throughput in synthetic benchmarks; in production, p99 produce latency dropped from recurring ∼500ms spikes to a stable ∼10ms (∼50×) and request-handler utilization fell from ∼80% to ∼20% – averting a projected nine-figure migration to an all-SSD broker fleet by proving the existing disk stack could meet I/O targets through optimization alone.
- **uServer:** Single-handedly built a self-service bare-metal provisioning platform adopted company-wide, managing hundreds of thousands of hosts across Uber's datacenters.
- Designed data security architecture and compliance tooling for global expansion into China, India, and Russia.

Rocket Lawyer San Francisco, CA
Senior Systems Engineer (original title: Systems Engineer) *2013 - 2015*

- Designed service infrastructure and provided 24/7 on-call support for a high-volume legal-services e-commerce site running primarily on the JVM and a legacy .NET platform. Improved reliability from 99% uptime to 99.99% and cut p99 latency from 2000ms to 150ms.

- Analyzed and optimized the performance of application services and the supporting infrastructure tier.
- Instrumented application code to expose internal state and key performance metrics, enrich logging, and reduce overall latency. Implemented dynamic service discovery and reconfiguration via ZooKeeper.
- Built tooling for continuous integration, dynamic configuration management, deployment management, monitoring, and automated incident detection and remediation.

nine.is Tuscaloosa, AL
Software Engineer / Web Developer *2012*

- Full-stack development and team leadership for a design company expanding into in-house software. Built applications using PHP, JavaScript, and Ruby on Rails.

The University of Alabama Tuscaloosa, AL
Graduate Research Assistant *2010 - 2012*

- Export-controlled research on cross-layer network optimization and security for next-generation internet infrastructure. Worked in the Linux networking stack – custom TCP-variant routing protocols and kernel-module development for fine-grained tuning and performance analysis – alongside an adaptive video-streaming client (gstreamer), automated VM-image penetration testing, and security analysis for the GENI experimental network testbed.

Brewer Porch Children's Center Tuscaloosa, AL
Unix Systems Administrator *2004 - 2006*

- Systems administration for a healthcare organization. Managed Linux/Solaris infrastructure and ensured HIPAA compliance for patient data.

PUBLICATIONS

- Dawei Li, Jason Bowman, Xiaoyan Hong "Evaluation of Security Vulnerabilities by Using ProtoGENI as a Launchpad", IEEE Globecom 2011, Houston, USA, Dec. 2011.
- M. Anderson, P. Kilgo, and J. Bowman. "RDIS: Generalizing domain concepts to specify device to framework mappings." In International Conference on Robotics and Automation, 2012. http://ua-robotics.net/index.php?title=RDIS

OUTREACH AND COMMUNITY INVOLVEMENT

- Mentor for Hackbright Academy, a 12 week intensive coding bootcamp focused on empowering women aiming to pursue a career in the software engineering field.
- Mentor for dev/Mission, a non-profit program that exposes low-income youth aged 16-24 to careers in technology.
- Organized and ran an introductory Scala and Functional Programming course based on Martin Oderskyś course and book. Resources available on my personal github.
- Served as president from 2010 - 2012 of the Association for Computing Machinery student chapter at The University of Alabama.
- Organized a mentoring/tutoring program for freshman and sophomore computer science students to improve program retention and graduation rates. Still going to this day, and graduation rates of freshman classes are up from 5% to nearly 50%.

INTERESTS

- **Professional:** Functional Programming · Language Development · Generative Programming · Operating Systems · System Architecture · Distributed Systems · Stream Processing · Virtualization · Open Source · GNU/Linux
- **Personal:** Reverse Engineering · Embedded Systems · Electrical Engineering · Robotics · Game Development & Theory · Music Production · Fabrication and DIY · CAD Design · Privacy · Mechanical Keyboards · Rock Climbing · Emacs